

LAB_07_Thomas_Travassos

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###LAB_07

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`

```
library(DBI)
library(RSQLite)

#Conexao com SQL

conn <- dbConnect(SQLite(), "datasets/database.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
SELECT * FROM subjects
WHERE abbreviation = 'STAT'
```

Table 1: 1 records

code	name	abbreviation
932	Statistics	STAT

```
SELECT DISTINCT instructors.id, instructors.name
FROM subjects
INNER JOIN subject_memberships
ON subjects.code = subject_memberships.subject_code
```

```

INNER JOIN sections
  ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name

```

Table 2: Displaying records 1 - 10

id	name
4787911	ABIGAIL BENZINE
3111842	ADIN-CRISTIAN ANDREI
3775809	AKICHIKA OZEKI
4958796	ALAN HUANG
5451706	ALBERTO DEL PIA
3661782	ALBRECHT KLEMM
5656034	ALEXANDER COVINGTON
4680082	ALEXANDER FISH
3573775	ALEXANDER KISELEV
5162339	ALYSSA DIGILIO

```

SELECT COUNT(DISTINCT instructors.id) AS numero_professores
FROM subjects
INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN sections
  ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'

```

Table 3: 1 records

numero_professores
379

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

-Quem é o professor mais difícil? -Quem é o professor mais fácil? -Qual é a disciplina mais difícil? -Qual é a disciplina mais fácil?

#Professor mais difícil;

```
WITH gpa_oferecimento AS (
  SELECT
    course_offering_uuid,
    (
      4.0 * SUM(a_count) +
      3.5 * SUM(ab_count) +
      3.0 * SUM(b_count) +
      2.5 * SUM(bc_count) +
      2.0 * SUM(c_count) +
      1.0 * SUM(d_count)
    ) /
    NULLIF(
      SUM(a_count) +
      SUM(ab_count) +
      SUM(b_count) +
      SUM(bc_count) +
      SUM(c_count) +
      SUM(d_count) +
      SUM(f_count),
      0
    ) AS GPA
  FROM grade_distributions
  GROUP BY course_offering_uuid
)

SELECT
  instructors.name,
  AVG(gpa_oferecimento.GPA) AS GPA_MEDIO
FROM gpa_oferecimento
INNER JOIN course_offerings
  ON gpa_oferecimento.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships
  ON course_offerings.uuid = subject_memberships.course_offering_uuid
```

```

INNER JOIN subjects
  ON subject_memberships.subject_code = subjects.code
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
  AND gpa_oferecimento.GPA IS NOT NULL
GROUP BY instructors.id, instructors.name
ORDER BY GPA_MEDIO ASC
LIMIT 1

```

Table 4: 1 records

name	GPA_MEDIO
JAMES D KUELBS	2.607692

#Professor mais facil;

```

WITH gpa_oferecimento AS (
  SELECT
    course_offering_uuid,
    (
      4.0 * SUM(a_count) +
      3.5 * SUM(ab_count) +
      3.0 * SUM(b_count) +
      2.5 * SUM(bc_count) +
      2.0 * SUM(c_count) +
      1.0 * SUM(d_count)
    ) /
    NULLIF(
      SUM(a_count) +
      SUM(ab_count) +
      SUM(b_count) +
      SUM(bc_count) +
      SUM(c_count) +
      SUM(d_count) +
      SUM(f_count),
      0
    )

```

```

    ) AS GPA
FROM grade_distributions
GROUP BY course_offering_uuid
)

SELECT
    instructors.name,
    AVG(gpa_oferecimento.GPA) AS GPA_MEDIO
FROM gpa_oferecimento
INNER JOIN course_offerings
    ON gpa_oferecimento.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
    ON subject_memberships.subject_code = subjects.code
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
    AND gpa_oferecimento.GPA IS NOT NULL
GROUP BY instructors.id, instructors.name
ORDER BY GPA_MEDIO DESC
LIMIT 1

```

Table 5: 1 records

name	GPA_MEDIO
SRIKANTHMADHAVAN ARAVAMUTHAN	4

#Disciplina mais dificil;

```

WITH gpa_oferecimento AS (
    SELECT
        course_offering_uuid,
        (

```

```

        4.0 * SUM(a_count) +
        3.5 * SUM(ab_count) +
        3.0 * SUM(b_count) +
        2.5 * SUM(bc_count) +
        2.0 * SUM(c_count) +
        1.0 * SUM(d_count)
    ) /
    NULLIF(
        SUM(a_count) +
        SUM(ab_count) +
        SUM(b_count) +
        SUM(bc_count) +
        SUM(c_count) +
        SUM(d_count) +
        SUM(f_count),
        0
    ) AS GPA
FROM grade_distributions
GROUP BY course_offering_uuid
)

SELECT
    courses.number,
    courses.name,
    AVG(gpa_oferecimento.GPA) AS GPA_MEDIO
FROM gpa_oferecimento
INNER JOIN course_offerings
    ON gpa_oferecimento.course_offering_uuid = course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
    ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
    AND gpa_oferecimento.GPA IS NOT NULL
GROUP BY courses.uuid, courses.number, courses.name
ORDER BY GPA_MEDIO ASC
LIMIT 1

```

Table 6: 1 records

number	name	GPA_MEDIO
431	Introduction to the Theory of Probability	2.87807

#Disciplina mais facil;

```

WITH gpa_oferecimento AS (
  SELECT
    course_offering_uuid,
    (
      4.0 * SUM(a_count) +
      3.5 * SUM(ab_count) +
      3.0 * SUM(b_count) +
      2.5 * SUM(bc_count) +
      2.0 * SUM(c_count) +
      1.0 * SUM(d_count)
    ) /
    NULLIF(
      SUM(a_count) +
      SUM(ab_count) +
      SUM(b_count) +
      SUM(bc_count) +
      SUM(c_count) +
      SUM(d_count) +
      SUM(f_count),
      0
    ) AS GPA
  FROM grade_distributions
  GROUP BY course_offering_uuid
)

SELECT
  courses.number,
  courses.name,
  AVG(gpa_oferecimento.GPA) AS GPA_MEDIO
FROM gpa_oferecimento
INNER JOIN course_offerings
  ON gpa_oferecimento.course_offering_uuid = course_offerings.uuid
INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid

```

```

INNER JOIN subject_memberships
  ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
  ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
  AND gpa_oferecimento.GPA IS NOT NULL
GROUP BY courses.uuid, courses.number, courses.name
ORDER BY GPA_MEDIO DESC
LIMIT 1

```

Table 7: 1 records

number	name	GPA_MEDIO
841	Nonparametric Statistics and Machine Learning Methods	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 20:16:44 -03"
```